

# CO4 AT1

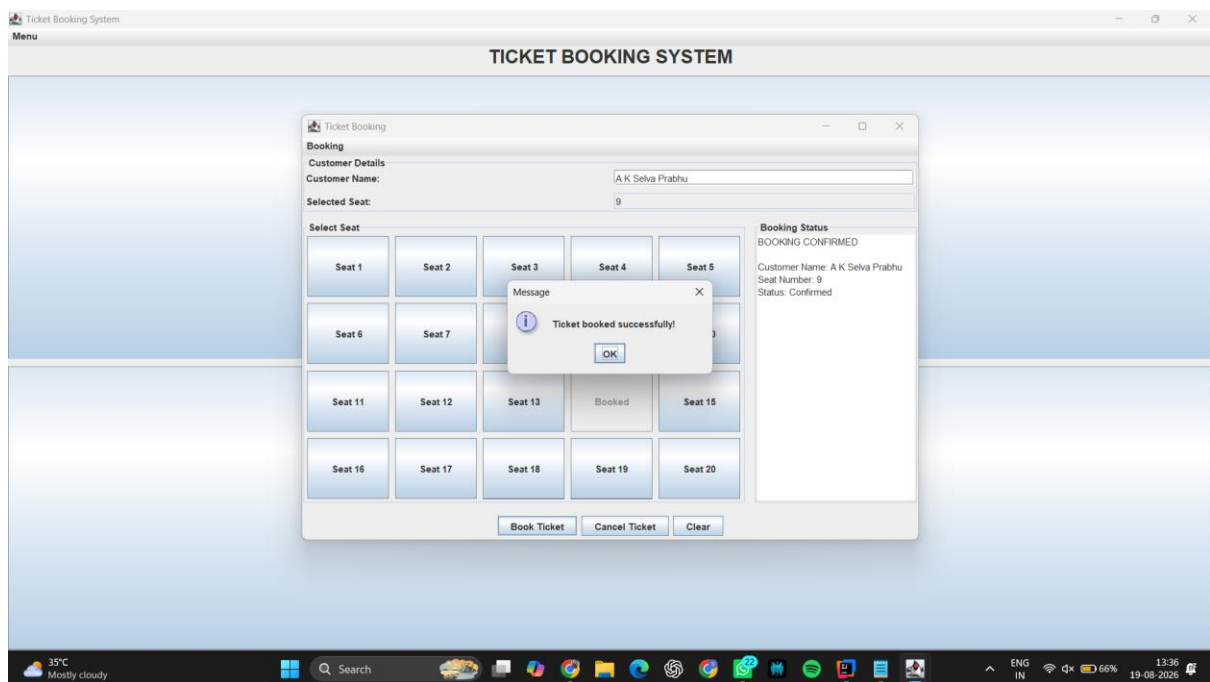
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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

## 1 Ticket Booking System



```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class TicketBookingSystem extends JFrame implements ActionListener {
```

```
    // Main window components
```

```
    JButton bookingWindowButton;
```

```
    JButton exitButton;
```

```
// Booking window components

JFrame bookingFrame;

JTextField nameField;

JTextField seatField;

JTextArea statusArea;


JButton[] seatButtons = new JButton[20];

boolean[] bookedSeats = new boolean[20];


JButton bookButton;

JButton cancelButton;

JButton clearButton;


public TicketBookingSystem() {

    // ----- MAIN WINDOW -----

    setTitle("Ticket Booking System");

    setSize(500, 300);

    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    setLocationRelativeTo(null);


    JPanel mainPanel = new JPanel(new BorderLayout(10, 10));

    JLabel title = new JLabel(

        "TICKET BOOKING SYSTEM",

        SwingConstants.CENTER

    );
```

```
title.setFont(new Font("Arial", Font.BOLD, 24));

mainPanel.add(title, BorderLayout.NORTH);

JPanel centerPanel = new JPanel(new GridLayout(2, 1, 10, 10));

bookingWindowButton =
    new JButton("Open Booking Window");

exitButton =
    new JButton("Exit");

bookingWindowButton.addActionListener(this);
exitButton.addActionListener(this);

centerPanel.add(bookingWindowButton);
centerPanel.add(exitButton);

mainPanel.add(centerPanel, BorderLayout.CENTER);

add(mainPanel);

createMenu();

setVisible(true);
}
```

```
// ----- MENU -----
```

```
private void createMenu() {
```

```
    JMenuBar menuBar = new JMenuBar();
```

```
    JMenu menu = new JMenu("Menu");
```

```
    JMenuItem openBooking =  
        new JMenuItem("Open Booking");
```

```
    JMenuItem exit =  
        new JMenuItem("Exit");
```

```
    openBooking.addActionListener(e -> openBookingWindow());
```

```
    exit.addActionListener(e -> System.exit(0));
```

```
    menu.add(openBooking);
```

```
    menu.addSeparator();
```

```
    menu.add(exit);
```

```
    menuBar.add(menu);
```

```
    setJMenuBar(menuBar);
```

```
}
```

```
// ----- BOOKING WINDOW -----
```

```
private void openBookingWindow() {

    if (bookingFrame != null && bookingFrame.isVisible()) {

        bookingFrame.toFront();

        return;

    }

    bookingFrame = new JFrame("Ticket Booking");

    bookingFrame.setSize(800, 550);

    bookingFrame.setLocationRelativeTo(this);

    bookingFrame.setLayout(new BorderLayout(10, 10));

    bookingFrame.setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

    // ----- CUSTOMER DETAILS -----

    JPanel detailsPanel =

        new JPanel(new GridLayout(2, 2, 10, 10));

    detailsPanel.setBorder(

        BorderFactory.createTitledBorder("Customer Details")

    );

    detailsPanel.add(new JLabel("Customer Name:"));

    nameField = new JTextField();

    detailsPanel.add(nameField);
```

```
detailsPanel.add(new JLabel("Selected Seat:"));
```

```
seatField = new JTextField();
```

```
seatField.setEditable(false);
```

```
detailsPanel.add(seatField);
```

```
bookingFrame.add(detailsPanel, BorderLayout.NORTH);
```

```
// ----- SEAT PANEL -----
```

```
JPanel seatPanel =
```

```
    new JPanel(new GridLayout(4, 5, 8, 8));
```

```
seatPanel.setBorder(
```

```
    BorderFactory.createTitledBorder("Select Seat")
```

```
);
```

```
for (int i = 0; i < 20; i++) {
```

```
    final int seatNumber = i + 1;
```

```
    seatButtons[i] =
```

```
        new JButton("Seat " + seatNumber);
```

```
    seatButtons[i].addActionListener(e -> {
```

```
        if (!bookedSeats[seatNumber - 1]) {
```

```
        seatField.setText(
            String.valueOf(seatNumber)
        );
    }
});

    seatPanel.add(seatButtons[i]);
}

bookingFrame.add(seatPanel, BorderLayout.CENTER);

// ----- STATUS AREA -----

statusArea = new JTextArea(8, 20);
statusArea.setEditable(false);

JScrollPane scrollPane =
    new JScrollPane(statusArea);

scrollPane.setBorder(
    BorderFactory.createTitledBorder("Booking Status")
);

bookingFrame.add(scrollPane, BorderLayout.EAST);

// ----- BUTTON PANEL -----

JPanel buttonPanel = new JPanel();
```

```
bookButton =  
    new JButton("Book Ticket");  
  
cancelButton =  
    new JButton("Cancel Ticket");  
  
clearButton =  
    new JButton("Clear");  
  
bookButton.addActionListener(this);  
cancelButton.addActionListener(this);  
clearButton.addActionListener(this);  
  
buttonPanel.add(bookButton);  
buttonPanel.add(cancelButton);  
buttonPanel.add(clearButton);  
  
bookingFrame.add(buttonPanel, BorderLayout.SOUTH);  
  
// Menu inside booking window  
createBookingMenu();  
  
bookingFrame.setVisible(true);  
}  
  
// ----- BOOKING MENU -----
```



```
private void createBookingMenu() {

    JMenuBar menuBar = new JMenuBar();

    JMenu bookingMenu =
        new JMenu("Booking");

    JMenuItem book =
        new JMenuItem("Book Ticket");

    JMenuItem cancel =
        new JMenuItem("Cancel Ticket");

    JMenuItem close =
        new JMenuItem("Close Window");

    book.addActionListener(e -> bookTicket());

    cancel.addActionListener(e -> cancelTicket());

    close.addActionListener(e -> bookingFrame.dispose());

    bookingMenu.add(book);
    bookingMenu.add(cancel);
    bookingMenu.addSeparator();
    bookingMenu.add(close);

    menuBar.add(bookingMenu);
}
```

```
        bookingFrame.setJMenuBar(menuBar);
    }

    // ----- EVENT HANDLING -----

    @Override
    public void actionPerformed(ActionEvent e) {

        if (e.getSource() == bookingWindowButton) {

            openBookingWindow();
        }

        else if (e.getSource() == exitButton) {

            System.exit(0);
        }

        else if (e.getSource() == bookButton) {

            bookTicket();
        }

        else if (e.getSource() == cancelButton) {

            cancelTicket();
        }
    }
}
```

```
else if (e.getSource() == clearButton) {

    clearFields();
}
}

// ----- BOOK TICKET -----

private void bookTicket() {

    try {

        String customerName =
            nameField.getText().trim();

        if (customerName.isEmpty()) {

            throw new Exception(
                "Please enter customer name."
            );
        }

        String seatText =
            seatField.getText().trim();

        if (seatText.isEmpty()) {
```

```
        throw new Exception(  
            "Please select a seat."  
        );  
    }
```

```
int seatNumber =  
    Integer.parseInt(seatText);
```

```
if (seatNumber < 1 || seatNumber > 20) {
```

```
    throw new Exception(  
        "Seat number must be between 1 and 20."  
    );  
}
```

```
if (bookedSeats[seatNumber - 1]) {
```

```
    throw new Exception(  
        "This seat is already booked."  
    );  
}
```

```
// Book the seat
```

```
bookedSeats[seatNumber - 1] = true;
```

```
seatButtons[seatNumber - 1]  
    .setText("Booked");
```

```

        seatButtons[seatNumber - 1]
            .setEnabled(false);

        statusArea.setText(
            "BOOKING CONFIRMED\n\n" +
            "Customer Name: " + customerName +
            "\nSeat Number: " + seatNumber +
            "\nStatus: Confirmed"
        );

        JOptionPane.showMessageDialog(
            bookingFrame,
            "Ticket booked successfully!"
        );
    }

    catch (NumberFormatException ex) {

        JOptionPane.showMessageDialog(
            bookingFrame,
            "Enter a valid seat number."
        );
    }

    catch (Exception ex) {

        JOptionPane.showMessageDialog(

```

```
        bookingFrame,
        ex.getMessage()
    );
}
}
```

```
// ----- CANCEL TICKET -----
```

```
private void cancelTicket() {

    try {

        String seatText =
            seatField.getText().trim();

        if (seatText.isEmpty()) {

            throw new Exception(
                "Please select a seat."
            );
        }

        int seatNumber =
            Integer.parseInt(seatText);

        if (seatNumber < 1 || seatNumber > 20) {

            throw new Exception(
```

```
        "Seat number must be between 1 and 20."
    );
}
```

```
if (!bookedSeats[seatNumber - 1]) {
```

```
    throw new Exception(
        "This seat is not booked."
    );
}
```

```
// Cancel booking
```

```
bookedSeats[seatNumber - 1] = false;
```

```
seatButtons[seatNumber - 1]
    .setText("Seat " + seatNumber);
```

```
seatButtons[seatNumber - 1]
    .setEnabled(true);
```

```
statusArea.setText(
    "BOOKING CANCELLED\n\n" +
    "Seat Number: " + seatNumber +
    "\nStatus: Cancelled"
);
```

```
JOptionPane.showMessageDialog(
    bookingFrame,
```

```

        "Ticket cancelled successfully!"
    );

}

catch (NumberFormatException ex) {

    JOptionPane.showMessageDialog(
        bookingFrame,
        "Enter a valid seat number."
    );
}

catch (Exception ex) {

    JOptionPane.showMessageDialog(
        bookingFrame,
        ex.getMessage()
    );
}
}

// ----- CLEAR -----

private void clearFields() {

    nameField.setText("");
    seatField.setText("");
}

```



```

        if (statusArea != null) {

            statusArea.setText("");

        }

    }

// ----- MAIN -----

public static void main(String[] args) {

    SwingUtilities.invokeLater(

        () -> new TicketBookingSystem()

    );

}

}

```

## 2 Customer Feedback System

**CUSTOMER FEEDBACK SYSTEM**

**Feedback Form**

Customer Name: A K Selva Prabhu

Email: akselvaprabhu@gmail.com

Rating: 1 - Very Poor

Feedback: good

**Submitted Feedback**  
 FEEDBACK SUBMITTED

Customer Name: A K Selva Prabhu  
 Email: akselvaprabhu@gmail.com  
 Rating: 1 - Very Poor  
 Feedback: good

```
import javax.swing.*;

import java.awt.*;

import java.awt.event.*;


public class CustomerFeedbackSystem extends JFrame

    implements ActionListener {


    JTextField nameField;

    JTextField emailField;


    JComboBox<String> ratingBox;


    JTextArea feedbackArea;

    JTextArea resultArea;


    JButton submitButton;

    JButton clearButton;


    public CustomerFeedbackSystem() {


        setTitle("Customer Feedback System");


        setSize(650, 500);


        setDefaultCloseOperation(

            JFrame.EXIT_ON_CLOSE

        );
```

```
setLocationRelativeTo(null);
```

```
setLayout(new BorderLayout(10, 10));
```

```
// ----- TITLE -----
```

```
JLabel title =
```

```
    new JLabel(  
        "CUSTOMER FEEDBACK SYSTEM",  
        SwingConstants.CENTER  
    );
```

```
title.setFont(  
    new Font("Arial", Font.BOLD, 24)  
);
```

```
add(title, BorderLayout.NORTH);
```

```
// ----- FORM -----
```

```
JPanel formPanel =
```

```
    new JPanel(  
        new GridLayout(4, 2, 10, 10)  
    );
```

```
formPanel.setBorder(  
    BorderFactory.createTitledBorder(  
        "CUSTOMER FEEDBACK SYSTEM"  
    )  
);
```

```
        "Feedback Form"

    )

);

// Name
formPanel.add(
    new JLabel("Customer Name:")
);

nameField = new JTextField();

formPanel.add(nameField);

// Email
formPanel.add(
    new JLabel("Email:")
);

emailField = new JTextField();

formPanel.add(emailField);

// Rating
formPanel.add(
    new JLabel("Rating:")
);

ratingBox =
```

```
new JComboBox<>(
    new String[]{
        "1 - Very Poor",
        "2 - Poor",
        "3 - Average",
        "4 - Good",
        "5 - Excellent"
    }
);

formPanel.add(ratingBox);

// Feedback
formPanel.add(
    new JLabel("Feedback:")
);

feedbackArea =
    new JTextArea(4, 20);

feedbackArea.setLineWrap(true);

feedbackArea.setWrapStyleWord(true);

formPanel.add(
    new JScrollPane(feedbackArea)
);
```

```
add(formPanel, BorderLayout.CENTER);
```

```
// ----- BUTTONS -----
```

```
JPanel buttonPanel =
```

```
    new JPanel();
```

```
submitButton =
```

```
    new JButton("Submit Feedback");
```

```
clearButton =
```

```
    new JButton("Clear");
```

```
submitButton.addActionListener(this);
```

```
clearButton.addActionListener(this);
```

```
buttonPanel.add(submitButton);
```

```
buttonPanel.add(clearButton);
```

```
add(buttonPanel, BorderLayout.SOUTH);
```

```
// ----- RESULT -----
```

```
resultArea =
```

```
    new JTextArea(8, 25);
```

```

resultArea.setEditable(false);

resultArea.setBorder(
    BorderFactory.createTitledBorder(
        "Submitted Feedback"
    )
);

add(
    new JScrollPane(resultArea),
    BorderLayout.EAST
);

setVisible(true);
}

// ----- EVENT HANDLING -----

@Override
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == submitButton) {

        submitFeedback();
    }

    else if (e.getSource() == clearButton) {

```

```
        clearForm();
    }
}

// ----- SUBMIT -----

private void submitFeedback() {

    try{

        String name =
            nameField.getText().trim();

        String email =
            emailField.getText().trim();

        String feedback =
            feedbackArea.getText().trim();

        // Name validation
        if (name.isEmpty()) {

            throw new Exception(
                "Please enter customer name."
            );
        }

        // Email validation
```



```
if (email.isEmpty()) {

    throw new Exception(

        "Please enter email."

    );
}

if (!email.contains("@")) {

    throw new Exception(

        "Please enter a valid email."

    );
}

// Feedback validation
if (feedback.isEmpty()) {

    throw new Exception(

        "Please enter your feedback."

    );
}

String rating =

    (String) ratingBox.getSelectedItemAt();

// Display result
resultArea.setText(

    "FEEDBACK SUBMITTED\n" +
```

```

        "-----\n\n" +
        "Customer Name: " + name +
        "\nEmail: " + email +
        "\nRating: " + rating +
        "\n\nFeedback:\n" + feedback
    );

    JOptionPane.showMessageDialog(
        this,
        "Feedback submitted successfully!"
    );
}

catch (Exception ex) {

    JOptionPane.showMessageDialog(
        this,
        ex.getMessage(),
        "Input Error",
        JOptionPane.ERROR_MESSAGE
    );
}
}

// ----- CLEAR -----

private void clearForm() {

```

```
nameField.setText("");
```

```
emailField.setText("");
```

```
ratingBox.setSelectedIndex(0);
```

```
feedbackArea.setText("");
```

```
resultArea.setText("");
```

```
}
```

```
// ----- MAIN -----
```

```
public static void main(String[] args) {
```

```
    SwingUtilities.invokeLater(
```

```
        () -> new CustomerFeedbackSystem()
```

```
    );
```

```
}
```

```
}
```